AI – IA 2

Agentic AI

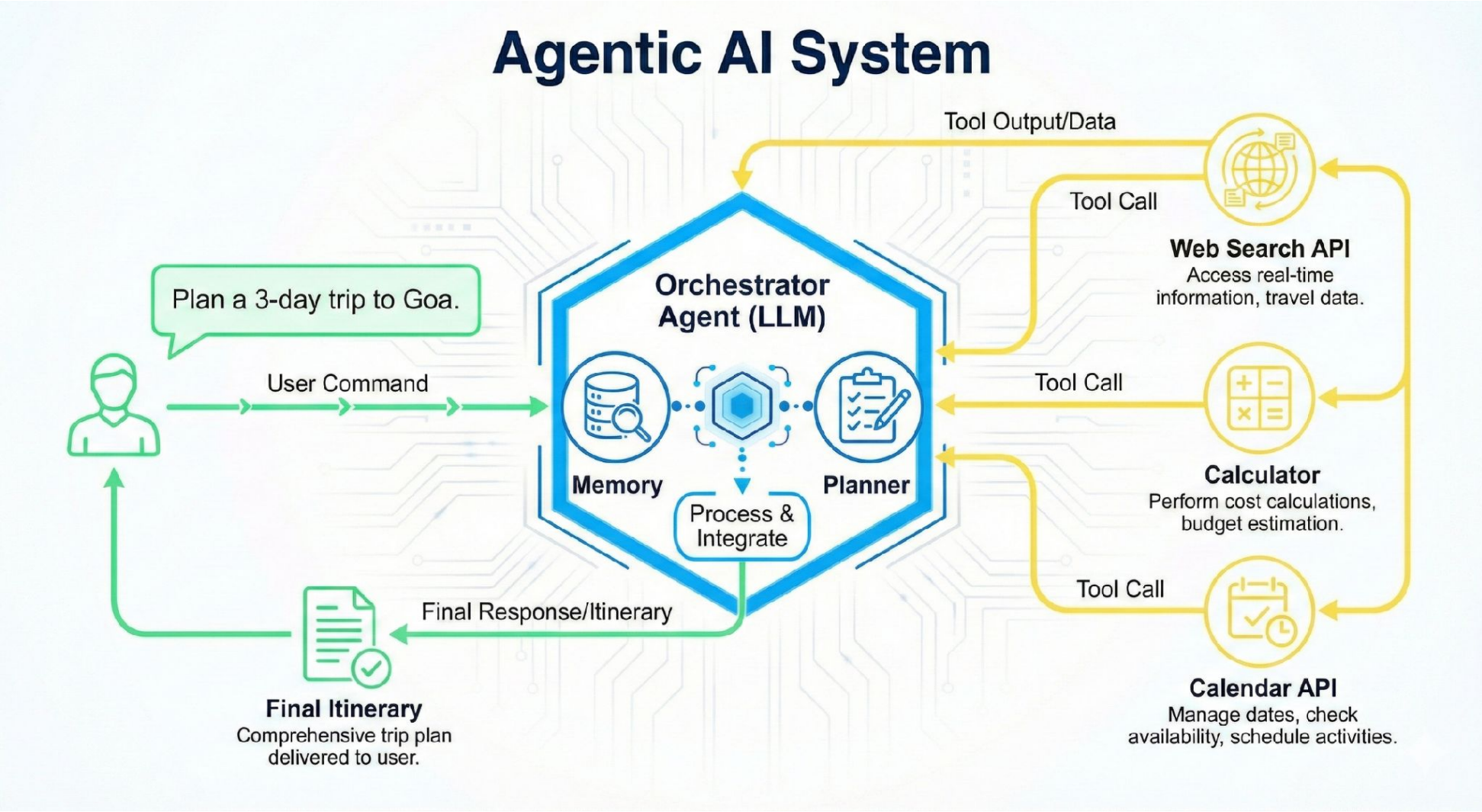
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Agentic AI: Extending Classical AI Agents

- In classical AI, we study **intelligent agents**
  - Perceive environment
  - Make decisions
  - Perform actions
- **Agentic AI** is a modern implementation of the same idea
- Difference:
  - Classical AI → theory & algorithms
  - Agentic AI → **Practical, real-world execution**
- Agentic AI focuses on:
  - Goal-based behavior
  - Planning & reasoning
  - Tool usage
  - Iterative decision making

- *Agentic AI = Classical AI agents + modern AI tools*

Agentic AI System Architecture



Agentic AI Workflow

1. **User Goal**
   - User provides a goal, not a single command
   - Example: “Plan a 3-day trip to Goa”
2. **Reasoning & Planning**
   - Agent breaks goal into sub-tasks
   - Decides sequence of actions
3. **Tool Use**
   - Agent uses tools when needed
   - Web search, calculator, calendar
4. **Loop & Result**
   - Agent evaluates results
   - Repeats planning if required
   - Produces final complete output

Agentic AI System Architecture

- **Orchestrator Agent (LLM)**
  - Acts as the controller
  - Coordinates reasoning and actions
- **Planner**
  - Decides what to do first, next, and last
- **Memory**
  - Stores intermediate results and context
- **Tool APIs**
  - Web Search → information
  - Calculator → numeric computation
  - Calendar → scheduling
- **Final Output**
  - Goal-complete response

Students Activity

In this activity, you will:

- Define a user goal
- Describe how the agent plans the task
- Identify which tools the agent uses
- Explain the loop and final output

AI Tools & Frameworks for Agentic AI Activity

- **Agent Frameworks (Choose ONE)**

1. **AutoGen**
   - Multi-agent conversation framework
   - Agents communicate to solve tasks
2. **CrewAI**
   - Role-based agents (Planner, Executor, Reviewer)
   - Clear task delegation
3. **ReAct-style Agents**
   - Reasoning + Action + Observation loop
   - Simple and well-aligned with AI syllabus
4. **BrowserUse**
   - Agent interacts with browser/webpages
   - Useful for research and data collection
5. **OpenRouter (Free API)**
   - Access to open-source LLMs
   - No paid subscription required
6. **LangChain**
   - Framework for building LLM-powered applications
   - Provides agent orchestration
   - Supports tool integration (APIs, search, databases)
   - Implements ReAct-style reasoning loop

AI Tools & Frameworks for Agentic AI Activity

- **Tools Integrated with Agents (Choose at least ONE)**

1. Web Search APIs
2. Calculator tools
3. Google OAuth 2.0 (Google Calendar tasks)
4. File / document readers (PDF, text)
5. Simple databases (CSV / JSON)
6. Python tools (basic scripting, data processing)

*Focus is on **agent behavior and reasoning**, not model training.*

Example Problem Statements

1. **Research Assistant Agent**
   - Reads documents (PDF / text files)
   - Uses tools to retrieve information
   - Answers user questions based on content
2. **Google Calendar Scheduling Agent**
   - Takes meeting details as input
   - Uses Google Calendar API
   - Automatically schedules the meeting
3. **Academic Planner Agent**
   - Creates a study schedule
   - Considers constraints (subjects, time, deadlines)
   - Adjusts plan dynamically
4. **Healthcare Information Agent (Non-Diagnostic)**
   - Collects user symptoms (text input)
   - Retrieves medical guidelines using web search
   - **Does NOT provide diagnosis**

Example Problem Statements

5. **Job Application Assistant Agent**
   - Analyzes Resume + Job Description
   - Identifies gaps
   - Suggests improvements
6. **Travel Planning Agent**
   - Plans itinerary for a trip
   - Uses web search for places, costs
   - Applies reasoning to optimize plan

*Each solution must demonstrate **Goal → Plan → Tool Use → Final Output**.*

"You are not building an AI model. You are designing an intelligent agent that can reason, plan, and use modern AI tools to achieve a goal."

References

- Agentic AI Tutorial for Beginners | Langgraph Tutorial
- Agents and Agentic Frameworks
- Best Agentic AI Framework
- Part 1 : ReACT AI Agents: A Guide to Smarter AI Through Reasoning and Action. | by Gauri | Medium
- Build Your First AI Agent in 20 Minutes (No Coding Required)